

BriefCase: Chain-of-Thought Distillation from an Open-Source Teacher for Legal Contract Risk Classification

Anuraag Kolli Ishaan Handa Mrigas Iyer Aaryan Srivastava
Purdue University

Abstract

Large language models (LLMs) can analyze legal contracts when prompted with chain-of-thought (CoT), but their inference cost and privacy implications make them impractical for large-scale contract review. We test whether training-time reasoning transfer from a locally-hosted open-weight teacher can substitute for in-domain pretraining in a compact encoder. Using CUAD, we generate structured rationales with Llama-3.1-8B-Instruct for every training clause, concatenate them to the clause at fine-tuning time, and remove them at inference. We evaluate a 2x2 design crossing encoder type (DistilBERT vs Legal-BERT) with training variant (vanilla vs CoT), plus direct teacher inference as a reference. CoT produces small positive effects on both students (+0.26 and +0.33 macro-F1 points) that fall inside single-seed noise. In-domain pretraining, by contrast, produces a robust 2-point gap that CoT does not close. The strongest finding is a ~ 47 -point gap between every fine-tuned student (≥ 0.87 macro-F1) and the teacher’s own zero-shot performance (0.40 macro-F1), which we argue bottlenecks the entire distillation pipeline: a teacher that cannot classify the task competently cannot transfer competent reasoning. We report per-category results showing that CoT is not uniformly weak, gaining +6 F1 on one category for both students, and discuss what these results imply for open-source CoT distillation in domain-specific NLP.

1 Introduction

Contract review is a time-intensive legal task: counsel must identify risky clauses, assess obligations, and flag unfavorable terms across hundreds of pages. LLMs can perform this analysis when prompted to reason step by step (Wei et al., 2022), but deploying LLMs at scale for contract review is expensive and raises data privacy concerns, since sensitive clauses often cannot leave client infrastructure.

Smaller encoders such as DistilBERT (Sanh et al., 2019) and Legal-BERT (Chalkidis et al., 2020) are cheap enough for production but lack the multi-step reasoning needed for nuanced legal judgment. Determining whether an indemnification clause is risky, for instance, requires understanding directionality (who indemnifies whom), scope limitations, and the presence or absence of liability caps. A keyword-matching model may flag “indemnify” without distinguishing protective from exploitative phrasing.

Chain-of-thought distillation (Hsieh et al., 2023; Magister et al., 2023) is a candidate remedy: a large teacher generates step-by-step rationales, a small student is fine-tuned on rationale-augmented inputs, and the rationale is dropped at inference. The hope is that exposure to structured reasoning at training time reshapes the student’s internal representations so it attends to the compositional features that matter, even without explicit reasoning at test time. Most prior CoT distillation work, however, uses proprietary API teachers, which is at odds with the privacy-sensitive deployment story that motivates legal NLP in the first place.

This project asks a focused question: *can training-time reasoning transfer from a local open-weight teacher substitute for in-domain pretraining in a compact encoder?* Concretely, we ask whether a generic DistilBERT trained with rationales from Llama-3.1-8B-Instruct can match a Legal-BERT trained on clauses alone. We implement a fully on-premise pipeline, run a 2x2 ablation over encoder type and training variant, and compare against direct zero-shot teacher inference. The entire pipeline runs in roughly 2.5 GPU-hours on a single A100-class GPU and costs \$0 in external API fees.

Our contributions are: (1) a reproducible open-weight pipeline for CoT distillation on CUAD, (2) a controlled 2x2 ablation isolating reasoning transfer from domain pretraining, (3) an honest negative result on the substitutability claim at 8B teacher

scale, and (4) a per-category analysis suggesting the primary bottleneck is teacher competence on the task, not the distillation framework itself.

2 Related Work

Knowledge distillation. Hinton et al. (2015) introduced the canonical formulation, where a student is trained to match the teacher’s softmax distribution under a temperature-scaled cross-entropy. Subsequent work has expanded this to intermediate representations and reasoning traces.

Chain-of-thought distillation. Hsieh et al. (2023) showed that task-specific rationales from LLMs can enable smaller models to outperform larger ones on several reasoning benchmarks, with less training data. Magister et al. (2023) obtained consistent gains on arithmetic and commonsense reasoning by fine-tuning small models on CoT-augmented corpora. Wang et al. (2023) proposed self-consistent CoT distillation to improve rationale faithfulness. Li et al. (2022) demonstrated LLM explanations improve small model NLU. West et al. (2022) showed symbolic knowledge distilled from GPT-3 can train commonsense models that exceed their teacher. All of this work uses proprietary teachers, and most targets general-purpose reasoning rather than closed-set domain classification.

Open-weight instruction-tuned LLMs. Recent open models have narrowed the gap with proprietary systems on reasoning benchmarks. Llama-3.1-8B-Instruct (Dubey et al., 2024) fits comfortably in a 40GB A100 in bfloat16 and is strong enough on general tasks to serve as a plausible distillation teacher, making it the natural candidate for a fully reproducible pipeline.

Legal NLP. CUAD (Hendrycks et al., 2021) provides 510 expert-annotated contracts across 41 clause types. Legal-BERT (Chalkidis et al., 2020) showed that legal-domain pretraining improves downstream legal NLP performance. LexGLUE (Chalkidis et al., 2022) established broader legal benchmarks. To our knowledge, prior work on CUAD has not tested whether reasoning transfer from open-weight LLMs can substitute for domain pretraining, which is the gap we address.

3 Method

3.1 Task formulation

Given an extracted contract clause c , we predict one of $K = 10$ risk-relevant categories derived from CUAD’s 41 clause types. We group related types into a reduced taxonomy to ensure each class has enough training examples for reliable fine-tuning and to simplify the downstream risk-assessment framing. Groupings were decided before any training was run and frozen thereafter. Otherwise, the annotated clauses in the database would be spread too thin per class to reliably train a classifier (see Table 1 for a reference on how we map 41 clause types to 10 categories).

3.2 Student encoders

We compare two encoders of similar scale and very different pretraining domains: **DistilBERT-base-uncased** (66M parameters, general-domain) and **Legal-BERT-base-uncased** (110M parameters, pretrained on legal corpora). Pairing these two against the vanilla-vs-CoT axis gives a clean 2x2 whose off-diagonal directly tests whether reasoning transfer substitutes for in-domain pretraining. We chose encoders rather than small decoder-only students because the task is closed-set classification, where [CLS]-based pooling with a linear head is a more natural fit than label-token generation, and because encoder inference is cheaper, matching the deployment scenario motivating the work.

3.3 Teacher model

We use **Llama-3.1-8B-Instruct** as the sole teacher, hosted locally via vLLM in bfloat16 on a single A100. This choice is deliberate: the model fits in 40GB with no quantization, rationale generation for the full CUAD training set completes in roughly 30 minutes, and no external service handles the contract text at any point. A larger teacher would likely produce higher-quality rationales; we discuss this trade-off extensively in §5 and leave teacher-scale ablations to future work.

3.4 Rationale generation

For each training clause c_i , we prompt the teacher with a fixed template requesting a four-step reasoning chain: (1) clause type, (2) favored party, (3) risky terms, (4) risk level (low / medium / high), capped at 100 output tokens. This produces rationales that explicitly surface directionality, scope, and risk-creating language, which are precisely

Table 1: Mapping of CUAD’s 41 clause types to the 10-class risk taxonomy used for our classification. Groupings follow legal risk theme, and collapsing to 10 categories ensures that we have sufficient per-class training examples from the 13k annotated clauses in CUAD. We have assigned index numbers to each risk category - these indices will be used to generate subsequent figures and will map to the categories they are associated with.

Risk Category	CUAD Clause Types
termination [index 0]	Termination For Convenience; Notice Period To Terminate Renewal; Effective Date; Expiration Date; Renewal Term
liability [index 1]	Cap On Liability; Uncapped Liability; Liquidated Damages; Insurance
ip_assignment [index 2]	IP Ownership Assignment; Joint IP Ownership; License Grant; Non-Transferable License; Affiliate License–Licensor; Affiliate License–Licensee; Unlimited/All-You-Can-Eat License; Irrevocable or Perpetual License; Source Code Escrow; Post-Termination Services
non_compete [index 3]	Non-Compete; Exclusivity; Competitive Restriction Exception; No-Solicit of Customers; No-Solicit of Employees; Non-Disparagement; ROFR/ROFO/ROFN
confidentiality [index 4]	Audit Rights; Covenant Not To Sue
payment [index 5]	Minimum Commitment; Volume Restriction; Price Restrictions; Most Favored Nation; Revenue/Profit Sharing
warranty [index 6]	Warranty Duration
change_of_control [index 7]	Change Of Control; Anti-Assignment
governing_law [index 8]	Governing Law
other [index 9]	Document Name; Parties; Agreement Date; Third Party Beneficiary

the compositional features we want the student to learn. Rationales are generated once and cached on disk by content hash so they are never regenerated across runs. Our exact prompt, verbatim, is : "Given the following contract clause, explain step by step: (1) what type of clause this is, (2) which party it favors, (3) what specific terms create risk, and (4) classify its risk level as low, medium, or high. Respond in under 100 tokens total. Clause: [c_i]"

3.5 Training and inference

Rationale-augmented training inputs are constructed as $\mathbf{x}_i^{train} = [\text{CLS}] c_i [\text{SEP}] r_i [\text{SEP}]$, while vanilla inputs use only $[\text{CLS}] c_i [\text{SEP}]$. All models are trained with standard cross-entropy on the classification head; the rationale has no dedicated loss term. At inference, **all** models receive only the clause, $\mathbf{x}_i^{test} = [\text{CLS}] c_i [\text{SEP}]$, so the CoT student must classify without any rationale. The hypothesis is that training-time exposure reshapes encoder representations in a way that generalizes without the crutch at test time.

3.6 Training details

All runs use identical hyperparameters: learning rate 2×10^{-5} with linear warmup over 10% of steps, batch size 16, max sequence length 512, AdamW with weight decay 0.01, up to 10 epochs

with early stopping on validation macro-F1 (patience 3), single seed (42). Contract-level 80/10/10 train/val/test splits prevent leakage. When a clause-plus-rationale input exceeds 512 tokens, the rationale is truncated first; the clause is preserved. The direct teacher baseline uses vLLM with bfloat16 weights, greedy decoding, and a labeled output parser that maps to one of the 10 risk categories or an `unknown` fallback.

Early stopping triggered for some of the four student runs (DistilBERT vanilla at epoch 8, DistilBERT+CoT at 10, Legal-BERT vanilla at 8, Legal-BERT+CoT at 9).

3.7 Deliberate scope decisions

Relative to the proposal, we made three scope reductions after preliminary analysis indicated the teacher had limited capability on the task and that additional runs would not change the qualitative conclusions: (1) we dropped the DeBERTa-v3 student, since the general-vs-legal contrast is already captured by DistilBERT-vs-Legal-BERT; (2) we dropped a second 32B teacher (Qwen2.5-32B), because the 8B results made teacher-scale comparison a higher-variance choice than a clean single-teacher story; (3) we dropped the Hinton-style logit-Knowledge Distillation baseline, as the CoT effect was already in the noise band and a separate KD arm would not have changed interpretation. Each

reduction is acknowledged as a limitation in §6.

4 Results

4.1 Experimental setup

We report test macro-F1 (primary), accuracy, and per-category F1 on a held-out contract-level split. All numbers are from a single seed (42); we treat any difference smaller than roughly 2 macro-F1 points as noise for interpretation purposes. End-to-end pipeline wall time on one A100 is approximately 2 hours 20 minutes: ~ 30 min for rationale generation, ~ 15 – 40 min per training run, and ~ 3 min for the teacher baseline.

The distribution is heavily skewed. The "other" class accounts for roughly 27% of training data because it catches all clause types outside the other risk categories, while warranty has only 131 training and 14 test examples. This skew is why we used two reporting choices. First, we use macro-averaged F1 as the primary metric so that small classes contribute equally to the headline number; raw accuracy would be dominated by other and ip_assignment, which together account for 44% of training data. Second, per-category F1 for warranty should be interpreted with caution because of its small amount of test examples ($n=14$), because a single misclassification shifts F1 by several points, so apparent CoT effects on this class are unlikely to be meaningful. We retain warranty in the macro-F1 average for completeness but flag it explicitly in per-category analysis.

Table 2: Per-class clause counts across train/val/test splits. Split is performed at the contract level (80/10/10) to prevent clause-level leakage.

Risk Category	Train	Val	Test
termination	1149	164	179
liability	1121	216	183
ip_assignment	1839	308	258
non_compete	1070	141	164
confidentiality	601	91	124
payment	804	138	136
warranty	131	31	14
change_of_control	678	110	119
governing_law	363	48	53
other	2824	379	387
Total	10,580	1,626	1,617

4.2 Main results

Table 3 summarizes test performance across the 2x2 plus the direct teacher. Figure 1 visualizes macro-F1 across all five conditions.

Student	Variant	Macro-F1	Acc.
DistilBERT	vanilla	0.8733	0.8821
DistilBERT	+ CoT	0.8759	0.8745
Legal-BERT	vanilla	0.8931	0.8967
Legal-BERT	+ CoT	0.8964	0.8929
Llama-3.1-8B 0-shot		0.4038	0.4062

Table 3: Main test-set results on CUAD with 10-class risk taxonomy. Best student result in bold. The teacher’s own zero-shot performance is ~ 47 macro-F1 points below every fine-tuned student. Computed over all 1,617 test examples; unparsed predictions counted as misclassifications. Macro-F1 over the 1,518 parsed predictions is 0.413; see Table 6.

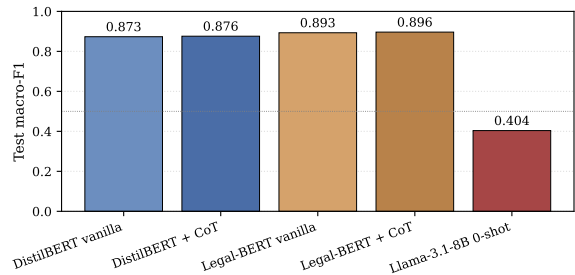


Figure 1: Test macro-F1 across all five conditions. All four fine-tuned students dominate the teacher’s direct zero-shot performance by a wide margin.

Three observations follow directly from Table 3. First, CoT provides a small positive gain on both students (+0.26 macro-F1 on DistilBERT, +0.33 on Legal-BERT), but both gains fall within our ~ 2 -point single-seed noise band. Second, in-domain pretraining produces a robust and larger effect: Legal-BERT outperforms DistilBERT by ~ 2 full F1 points regardless of CoT variant. Third, and most striking, **every** fine-tuned student outperforms direct zero-shot inference from its own teacher by roughly 47 macro-F1 points.

4.3 Does CoT close the domain-pretraining gap?

No. The key off-diagonal comparison is DistilBERT+CoT (0.8759) versus Legal-BERT vanilla (0.8931), a gap of 1.7 macro-F1 points. If reasoning transfer could substitute for domain pretraining, we would expect this gap to close or reverse. It does not. The vertical (vanilla-to-CoT) delta is an order of magnitude smaller than the horizontal (DistilBERT-to-Legal-BERT) delta across the 2x2, so at 8B teacher scale, rationale augmentation is not a substitute for a legal pretraining corpus.

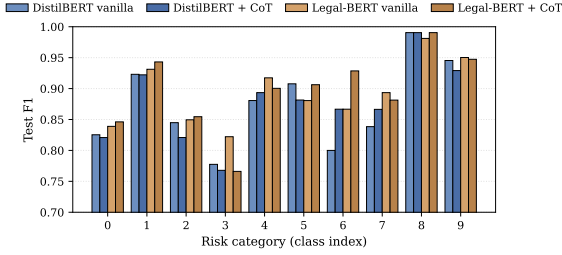


Figure 2: Per-category test F1. All 2x2 cells are shown; category 8 is easy for every condition, while categories 3 and 6 separate the vanilla-vs-CoT variants most visibly.

4.4 Per-category analysis

Figures 2 and 3 disaggregate performance by risk category. While aggregate CoT gains are small, per-category effects are heterogeneous. warranty (index 6) is the best positive case: both students gain approximately 6 F1 points from CoT (+6.7 for DistilBERT, +6.2 for Legal-BERT), well outside the single-seed noise band — though, as flagged in 4.1, with only 14 test examples this corresponds to roughly one fewer misclassification. In contrast, non_compete (index 3) loses 5.6 F1 points on Legal-BERT when CoT is added, the single largest negative effect in the experiment. confidentiality and change_of_control (indices 4 and 7) show mixed signs across students. The asymmetry maps cleanly onto Table 6. Warranty is the only class on which the teacher achieves perfect recall (1.000), while the teacher’s recall on non_compete is just 0.360 which means that roughly 64% of true non_compete training examples received rationales with incorrect step-(4) labels. The confusion matrix (Table 4) makes the failure mode concrete: of 161 true non_compete clauses, the teacher routes 33 to termination, 22 to payment, and 16 to governing_law. Categories where the teacher’s zero-shot predictions align with the gold distribution see positive CoT transfer; categories where the teacher is systematically wrong see degradation.

4.5 Direct teacher performance and Student sizes/performances

the teacher’s macro-F1 of 0.4038 is 47 points below the weakest fine-tuned student. By comparison, a trivial majority-class predictor (always other) would achieve macro-F1 is roughly 0.04. The unknown-label rate is 6%, indicating that most failures are confidently wrong rather than unparseable. This is very relevant: the teacher produces the rationales the student learns from, and a teacher that is

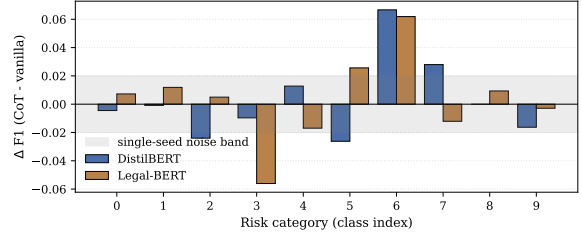


Figure 3: Per-category $\Delta F1$ (CoT minus vanilla) for each student. The shaded band represents the single-seed noise region (± 0.02). Category 6 shows a large positive effect on both students; category 3 shows the largest negative effect on Legal-BERT.

correct on only 40% of training clauses will embed incorrect step-(4) conclusions in a corresponding fraction of rationales.

Table 5 shows the breakdown of the sizes of each student model. The four student models are compact and fast at inference. DistilBERT (66.4M parameters) processes each clause in approximately 1.2 ms, while Legal-BERT (109.5M parameters) takes roughly 2.1 ms — nearly twice as slow that scales predictably with the $1.65\times$ parameter ratio between the two architectures. Crucially, the CoT and vanilla variants of each model are effectively identical in both size and latency, since rationales are present only during training and are stripped at test time. This means the CoT augmentation improves classification quality at zero inference cost, making it a particularly attractive technique for latency-sensitive legal NLP deployments.

Table 4: Confusion matrix for the Llama-3.1-8B zero-shot baseline (rows = true label, columns = predicted label).

	term	liab	ip	ncomp	conf	pay	warr	chg	gov	oth
term	122	0	0	0	2	2	0	2	46	5
liab	9	109	0	0	0	42	17	1	0	1
ip	39	5	35	6	5	66	8	3	28	4
ncomp	33	3	5	58	7	22	1	11	16	5
conf	8	10	6	1	14	32	11	0	32	3
pay	8	1	0	0	0	116	9	0	1	1
warr	0	0	0	0	0	0	14	0	0	0
chg	23	2	31	8	0	1	1	35	2	1
gov	0	0	0	0	0	0	0	0	53	0
oth	0	1	0	0	0	36	1	0	253	85

5 Analysis and Discussion

5.1 Why did CoT not substitute for domain pretraining?

The clearest explanation is that the teacher itself is the bottleneck. Rationale quality is upper-bounded

Table 5: Model size and inference latency on the test set (1,617 clauses). CoT and vanilla variants are identical at inference since rationales are stripped at test time.

Model	Params (M)	ms / sample	Test runtime (s)
DistilBERT vanilla	66.4	1.224	1.93
DistilBERT + CoT	66.4	1.250	1.97
Legal-BERT vanilla	109.5	2.135	3.37
Legal-BERT + CoT	109.5	2.136	3.37

Table 6: Per-class precision, recall, and F1 for the Llama-3.1-8B zero-shot baseline on the test set (1,578 parsed predictions; 6% unparsed outputs excluded) due to the model producing a response outside of the 10 risk categories for 99 clauses. The macro-F1 was computed over the 1518 valid predictions as a result. Per-class supports differ from Table 2 because unparsed teacher predictions are excluded; the difference per class equals the number of unparsed predictions for that class

Risk Category	Precision	Recall	F1	Support
termination	0.504	0.682	0.580	179
liability	0.832	0.609	0.703	179
ip_assignment	0.455	0.176	0.254	199
non_compete	0.795	0.360	0.496	161
confidentiality	0.500	0.120	0.193	117
payment	0.366	0.853	0.512	136
warranty	0.226	1.000	0.368	14
change_of_control	0.673	0.337	0.449	104
governing_law	0.123	1.000	0.219	53
other	0.810	0.226	0.353	376
Macro avg	0.528	0.536	0.413	1,518

by the teacher’s task understanding: a teacher that misclassifies clauses will reason toward those incorrect classifications. The teacher’s zero-shot accuracy of 40% (as shown in Table 3) implies that nearly 60% of training rationales embed an incorrect step-(4) judgment in otherwise fluent legal prose. The confusion matrix (Table 4) makes this concrete: the teacher routes 253 other clauses to governing_law, 66 ip_assignment clauses to payment, and 42 liability clauses to payment, so the rationales generated for those clauses during training carry confidently wrong final classifications. According to official documentation, Llama-3.1-8B-Instruct was trained on general web data with some legal content but no explicit legal-classification objective, and at 8B parameters it has limited capacity for the fine distinctions which CUAD’s taxonomy likely requires. The fact that CoT still produced small positive effects, rather than net-negative effects, suggests the student was learning from the structural content of rationales (what to attend to) even when the conclusional content was wrong. This interpretation is consistent with the heterogeneous per-category effects in 4.4: CoT helps on classes where the teacher’s per-class recall is high (warranty: 1.000, Table 6) and hurts on classes where the teacher’s recall is low (non_compete:

0.360), tracking exactly the pattern the bottleneck account predicts.

5.2 Why did CoT help the Warranty category on both students?

Mapping back to Table 1, category 6 corresponds to the warranty class. We can now resolve our two original hypotheses. The deceptive-surface-form hypothesis does not fit: warranty clauses are typically self-identifying through explicit warranty language. The teacher-accuracy hypothesis does fit, and strongly: the teacher’s zero-shot recall on warranty is 1.000 (Table 6), well above its macro-averaged recall of 0.536. Every true warranty clause in the test set is correctly identified by the teacher, so the rationales generated for warranty training clauses carry correct step-(4) judgments, which is precisely the regime in which the 5.1 bottleneck argument predicts CoT should help.

However, two caveats apply. First, the teacher’s precision on warranty is only 0.226, meaning it over-predicts the class broadly; the perfect recall reflects a "default to warranty when uncertain" tendency more than discriminative skill. The rationales generated for true warranty clauses are accurate at step 4, but the teacher’s overall handling of the class is noisier than recall alone suggests.

Second, with only 14 warranty test examples, the +6 F1 gain corresponds to roughly one fewer misclassification. The qualitative pattern is consistent with our bottleneck account, but the warranty result alone should not be over-weighted as evidence for CoT efficacy.

5.3 Teacher-vs-student gap

The ~ 47 -point teacher-vs-student gap is, in our view, the most externally important finding of this work. It says that for a closed-set legal classification task with expert labels, a small fine-tuned encoder is radically more reliable than direct LLM inference from a mid-size open-weight model. A practitioner faced with the choice between deploying a 110M-parameter Legal-BERT fine-tune at 0.89 macro-F1 or an 8B LLM at 0.40 macro-F1 has a clear answer. This also clarifies when CoT distillation would be expected to help: it requires a teacher whose zero-shot performance is already close to or above what a fine-tuned student can achieve. At 8B parameters, that condition is not met for this task.

5.4 Practical takeaways

Three takeaways generalize beyond this project. (1) CoT distillation is only as strong as the teacher’s understanding of the downstream task; teacher-scale is a first-class design parameter. (2) For closed-set classification in specialized domains, fine-tuned encoders deliver around $2\times$ the macro-F1 at roughly 1/100 the parameter count and millisecond-scale latency (Table 5). (3) Open-weight, on-premise pipelines can fully reproduce every part of a CoT distillation study at near-zero cost, which we think should become the default in legal and other privacy-sensitive NLP settings.

6 Limitations

Single seed. All numbers are from seed 42. Differences smaller than ~ 2 macro-F1 points should not be interpreted as significant. We estimate that three-seed error bars would add ~ 3 GPU-hours; we prioritized breadth of comparison over seed coverage.

Single teacher. With only Llama-3.1-8B, we cannot test whether a larger or domain-adapted teacher would change the reasoning-vs-pretraining trade-off. The prediction that follows from our analysis is that a strong enough teacher would swing the

outcome, but this is currently a prediction and not a measurement.

No logit-KD baseline. We do not compare against Hinton-style logit distillation in this report, so we cannot cleanly attribute the weak CoT gains to rationale content versus the generic regularization effect of seeing additional training-time tokens.

Taxonomy reduction. Grouping CUAD’s 41 clause types into 10 risk categories was a subjective step. We fixed the mapping before training and did not iterate on it, but alternative groupings could plausibly shift per-category results.

Dataset scope. CUAD is English-language, U.S.-jurisdiction commercial contracts only. Generalization to other legal systems or document types is untested.

Rationale quality. We did not conduct a manual audit of rationale correctness. The teacher’s 40% zero-shot accuracy implies that a substantial fraction of step-(4) judgments in training rationales are wrong; a formal audit would strengthen the bottleneck claim.

7 Conclusion

We tested whether chain-of-thought distillation from an open-weight teacher can substitute for in-domain pretraining in a small legal-clause classifier. The answer, at 8B teacher scale on CUAD, is no: CoT produces effects inside the single-seed noise band, while domain pretraining produces a robust 2-point gap that CoT does not close. The strongest result in our data is the ~ 47 -point advantage of every fine-tuned student over direct zero-shot teacher inference, which both justifies the distillation framing and identifies teacher competence on the task as the rate-limiting factor for reasoning transfer. Future work should scale the teacher and audit rationale correctness; our pipeline is reproducible end-to-end in 2.5 GPU-hours at zero external cost, and all code and results will be released alongside this report.

References

Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. LEGAL-BERT: The muppets straight out of law school. In *Findings of EMNLP*.

- Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Martin Katz, and Nikolaos Aletras. 2022. LexGLUE: A benchmark dataset for legal language understanding in english. In *Proceedings of ACL*.
- Abhimanyu Dubey and 1 others. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.
- Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. 2021. CUAD: An expert-annotated NLP dataset for legal contract review. In *Proceedings of NeurIPS Datasets and Benchmarks*.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. In *NeurIPS Deep Learning Workshop*.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of ACL*.
- Shiyang Li, Jianshu Chen, Yelong Shen, Zhoujun Chen, Xinlu Zhang, Zekun Li, Hong Wang, Jing Qian, Baolin Peng, Yi Mao, Wenhui Chen, and Xifeng Yan. 2022. Explanations from large language models make small reasoners better. *arXiv preprint arXiv:2210.06726*.
- Lucie Charlotte Magister, Jonathan Mallinson, Jakub Adamek, Eric Malmi, and Aliaksei Severyn. 2023. Teaching small language models to reason. In *Proceedings of ACL*.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. In *NeurIPS EMC² Workshop*.
- Peifeng Wang, Zhengyang Wang, Zheng Li, Yifan Gao, Bing Yin, and Xiang Ren. 2023. SCOTT: Self-consistent chain-of-thought distillation. In *Proceedings of ACL*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Peter West, Chandra Bhagavatula, Jack Hessel, Jena D. Hwang, Liwei Jiang, Ronan Le Bras, Ximing Lu, Sean Welleck, and Yejin Choi. 2022. Symbolic knowledge distillation: from general language models to commonsense models. In *Proceedings of NAACL*.